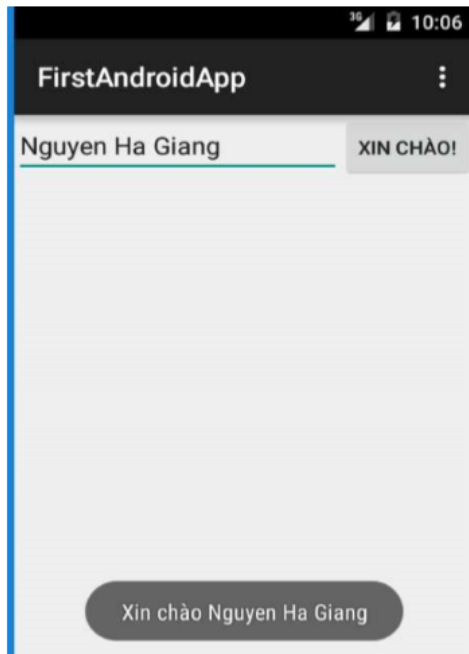
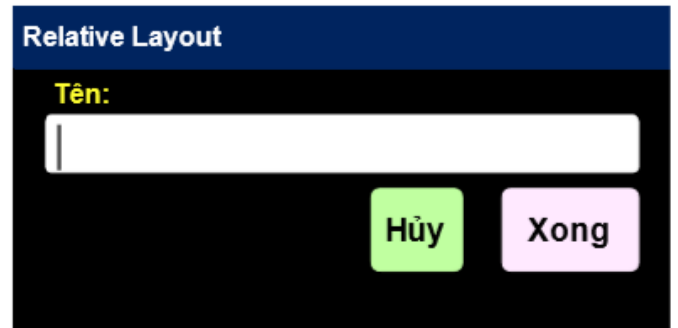


Lab 0 – Layout cơ bản

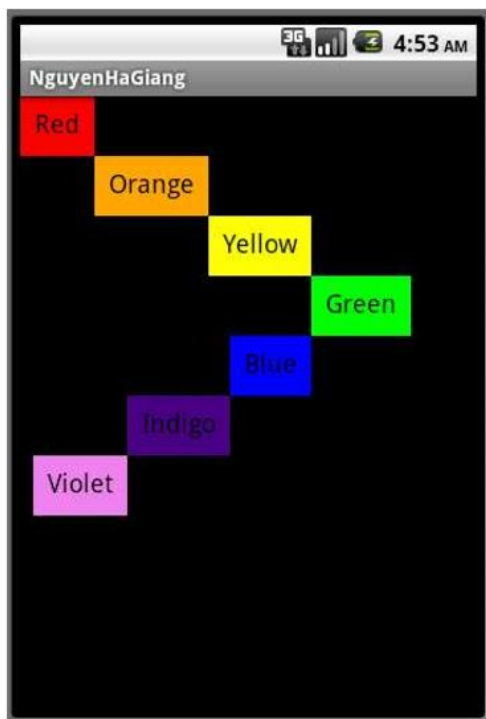
Tạo project và thiết kế các màn hình như sau:



Hình 1



Hình 2



Hình 3



Hình 4

Đăng nhập vào hệ thống

CSCOM

TenDangNhap

Đăng nhập

☒ Lưu thông tin đăng nhập

Annotations: 1 points to the title bar, 2 to the logo, 3 to the username field, 4 to the password field, 5 to the login button, and 6 to the remember me checkbox.

Hình 5

CẬP NHẬT THÔNG TIN CÁ NHÂN

1 CMND:

Số điện thoại:

Địa chỉ:

2

3

4 Cập nhật Làm lại

Annotation 1 points to the ID card label, 2 to the ID card input field, 3 to the profile picture, and 4 to the update and reset buttons.

Hình 6

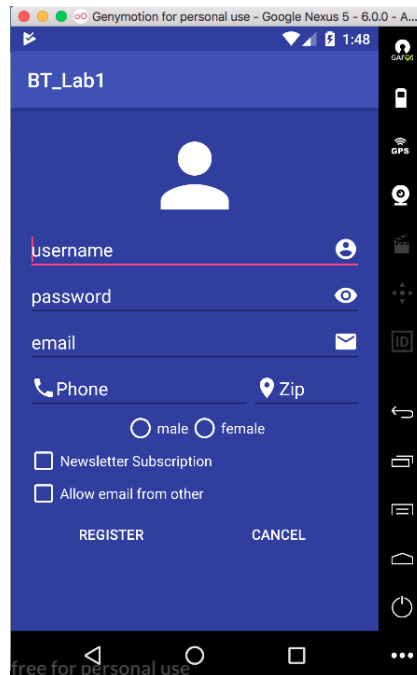
Lab 1: TextView, EditText, Button , ImageView

Sau khi thực hiện bài Lab này sinh viên:

- Triển khai ứng dụng với View cơ bản như TextView, EditText, Button, ImageView
- Vận dụng các thuộc tính cơ bản

Yêu cầu:

- Thiết kế giao diện như sau



Gợi ý:

Phần thiết kế giao diện:

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:background="@color/colorPrimaryDark"
    android:padding="16dp"
    tools:context="com.example.vudinhai.bt_project1.MainActivity">

    <ImageView
        android:layout_width="100dp"
        android:layout_height="100dp"
        android:src="@drawable/ic_person_black_24dp"
        android:layout_gravity="center"/>

    <EditText
        android:layout_width="match_parent"
```

```

        android:layout_height="wrap_content"
        android:hint="username"
        android:id="@+id/edtUsername"
        android:textColorHint="#FFFFFF"
        android:drawableEnd="@drawable/ic_account_circle_black_24dp"/>

<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="password"
    android:id="@+id/edtPassword"
    android:textColorHint="#FFFFFF"
    android:drawableEnd="@drawable/ic_visibility_black_24dp"/>
<EditText
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="email"
    android:id="@+id/edtEmail"
    android:textColorHint="#FFFFFF"
    android:drawableEnd="@drawable/ic_mail_black_24dp"/>

<LinearLayout
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">

    <EditText
        android:layout_width="0dp"
        android:layout_weight="2"
        android:layout_height="wrap_content"
        android:drawableLeft="@drawable/ic_local_phone_black_24dp"
        android:hint="Phone"
        android:id="@+id/edtPhone"
        android:textColorHint="#FFFFFF" />

    <EditText
        android:layout_width="0dp"
        android:layout_weight="1"
        android:layout_height="wrap_content"
        android:drawableLeft="@drawable/ic_location_on_black_24dp"
        android:hint="Zip"
        android:id="@+id/edtZip"
        android:textColorHint="#FFFFFF" />

</LinearLayout>

```

```

<RadioGroup
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal"
    android:gravity="center"
    android:id="@+id/rdg">

    <RadioButton
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:id="@+id/rdb1"
        android:text="male"
        android:textColor="#FFFFFF"
        android:buttonTint="#FFFFFF"
    />

    <RadioButton
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="female"
        android:id="@+id/rdb2"
        android:textColor="#FFFFFF"
        android:buttonTint="#FFFFFF"
    />

</RadioGroup>

<CheckBox
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Newsletter Subscription"
    android:id="@+id/chk1"
    android:textColor="#FFFFFF"
    android:buttonTint="#FFFFFF"
/>

<CheckBox
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Allow email from other"
    android:id="@+id/chk2"
    android:textColor="#FFFFFF"
    android:buttonTint="#FFFFFF"
/>
<LinearLayout
    android:layout_width="match_parent"

```

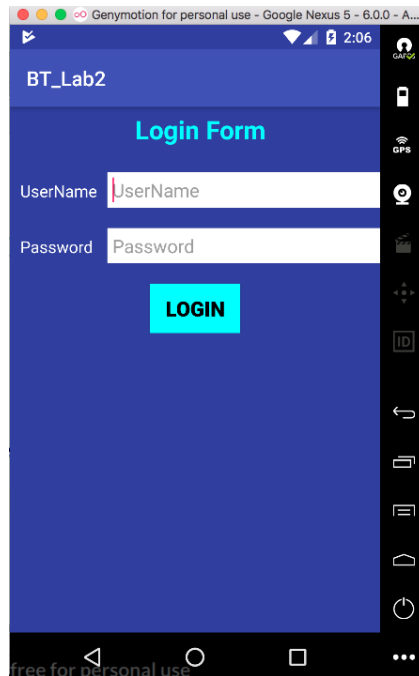
```
        android:layout_height="wrap_content"
        android:orientation="horizontal">
        <Button
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:text="Register"
            android:id="@+id/btn1"
            android:textColor="#FFFFFF"
            android:onClick="onClickXML"
            android:background="@color/colorPrimaryDark"/>

        <Button
            android:layout_width="0dp"
            android:layout_weight="1"
            android:layout_height="wrap_content"
            android:text="Cancel"
            android:id="@+id/btn2"
            android:onClick="onClickXML"
            android:textColor="#FFFFFF"
            android:background="@color/colorPrimaryDark"/>
    </LinearLayout>

</LinearLayout>
```

Lab 2 : Xây dựng màn hình Login với TableLayout

Yêu cầu : Xây dựng giao diện sau



Hướng dẫn:

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<TableLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:background="@color/colorPrimaryDark"  
    android:orientation="vertical"  
    android:stretchColumns="1">
```

```
    <TableRow android:padding="5dp">
```

```
        <TextView
```

```
            android:layout_height="wrap_content"  
            android:layout_marginBottom="20dp"  
            android:layout_span="2"  
            android:gravity="center_horizontal"  
            android:text="@string/loginForm"  
            android:textColor="#0ff"  
            android:textSize="25sp"  
            android:textStyle="bold" />
```

```
</TableRow>
```

```
<TableRow>
```

```
<TextView
    android:layout_height="wrap_content"
    android:layout_column="0"
    android:layout_marginLeft="10dp"
    android:text="@string/userName"
    android:textColor="#fff"
    android:textSize="16sp" />
```

```
<EditText
    android:id="@+id/userName"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_marginLeft="10dp"
    android:background="#fff"
    android:hint="@string/userName"
    android:padding="5dp"
    android:textColor="#000" />
```

```
</TableRow>
```

```
<TableRow>
```

```
<TextView
    android:layout_height="wrap_content"
    android:layout_column="0"
    android:layout_marginLeft="10dp"
    android:layout_marginTop="20dp"
    android:text="@string/password"
    android:textColor="#fff"
    android:textSize="16sp" />
```

```
<EditText
    android:id="@+id/password"
    android:layout_height="wrap_content"
    android:layout_column="1"
    android:layout_marginLeft="10dp"
    android:layout_marginTop="20dp"
    android:background="#fff"
    android:hint="@string/password"
    android:padding="5dp"
    android:textColor="#000" />
```

```
</TableRow>
```



```
<TableRow android:layout_marginTop="20dp">
```

```
    <Button
```

```
        android:id="@+id/loginBtn"
```

```
        android:layout_height="wrap_content"
```

```
        android:layout_gravity="center"
```

```
        android:layout_span="2"
```

```
        android:background="#0ff"
```

```
        android:text="@string/login"
```

```
        android:textColor="#000"
```

```
        android:textSize="20sp"
```

```
        android:textStyle="bold" />
```

```
</TableRow>
```

values/strings

```
<resources>
```

```
    <string name="app_name">BT_Lab2</string>
```

```
    <string name="hello_world">Hello world!</string>
```

```
    <string name="action_settings">Settings</string>
```

```
    <string name="loginForm">Login Form</string>
```

```
    <string name="userName">UserName</string>
```

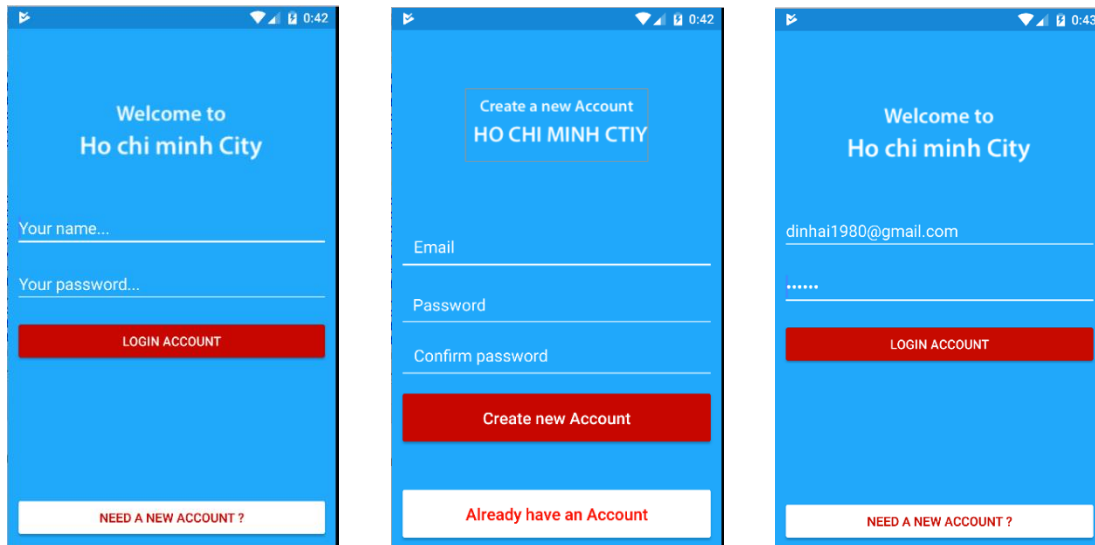
```
    <string name="password">Password</string>
```

```
    <string name="login">LogIn</string>
```

```
</resources>
```

Lab 3: Thiết kế giao diện với Constraintlayout và các view cơ bản

Yêu cầu : thiết kế giao diện như hình sau



Hướng dẫn code:

Bước 1: Xây dựng theme cho giao diện

Vào res/values/styles.xml

<resources>

```
<!-- Base application theme. -->
<style name="AppTheme" parent="Theme.AppCompat.Light.NoActionBar">
    <!-- Customize your theme here. -->
    <item name="colorPrimary">@color/colorPrimary</item>
    <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
    <item name="colorAccent">@color/colorAccent</item>
</style>
```

</resources>

Bước 2: Xây dựng Activity Login

Right chuột vào thư mục JAVA → New → Activity → Empty Activity → Đặt tên LoginActivity

<?xml version="1.0" encoding="utf-8"?>

<android.support.constraint.ConstraintLayout

xmlns:android="http://schemas.android.com/apk/res/android"

xmlns:app="http://schemas.android.com/apk/res-auto"

xmlns:tools="http://schemas.android.com/tools"

android:layout_width="match_parent"

android:layout_height="match_parent"

tools:context=".LoginActivity"

android:background="@color/colorPrimary">

<ImageView

android:id="@+id/imageView"

android:layout_width="200dp"

```

    android:layout_height="100dp"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="64dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:srcCompat="@drawable/logo" />

```

<EditText

```

    android:id="@+id/editText"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="32dp"
    android:ems="10"
    android:inputType="textPersonName"
    android:hint="Your name..."
    android:textColorHint="#FFFFFF"
    android:textColor="#FFFFFF"
    android:backgroundTint="@android:color/white"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintHorizontal_bias="0.503"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/imageView" />

```

<EditText

```

    android:id="@+id/editText2"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:ems="10"
    android:inputType="textPassword"
    android:hint="Your password..."
    android:textColorHint="#FFFFFF"
    android:textColor="#FFFFFF"
    android:backgroundTint="@android:color/white"
    app:layout_constraintEnd_toEndOf="@+id/editText"
    app:layout_constraintHorizontal_bias="0.0"
    app:layout_constraintStart_toStartOf="@+id/editText"
    app:layout_constraintTop_toBottomOf="@+id/editText" />

```

<Button

```

    android:id="@+id/button"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="Login account"
    android:backgroundTint="@android:color/holo_red_dark"
    android:textColor="#FFFFFF"
    app:layout_constraintEnd_toEndOf="@+id/editText2"
    app:layout_constraintStart_toStartOf="@+id/editText2"

```

```
app:layout_constraintTop_toBottomOf="@+id/editText2" />
```

```
<Button
```

```
    android:id="@+id/button2"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginBottom="8dp"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:textColor="@android:color/holo_red_dark"
    android:backgroundTint="@android:color/white"
    android:text="Need a new account ?"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent" />
```

```
</android.support.constraint.ConstraintLayout>
```

Bước 3: Xây dựng activity register thao tác tương tự như Login → đặt tên RegisterActivity

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<android.support.constraint.ConstraintLayout
```

```
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".RegisterActivity"
    android:background="@color/colorPrimary">
```

```
<ImageView
```

```
    android:id="@+id/regPageLogo"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="64dp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent"
    app:srcCompat="@drawable/logo2" />
```

```
<EditText
```

```
    android:id="@+id/reg_email"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="64dp"
    android:backgroundTint="@android:color/white"
    android:ems="10"
    android:hint="Email"
    android:inputType="textEmailAddress"
    android:padding="16dp"
    android:textColor="@android:color/white"
```

```

    android:textColorHint="#FFFFFF"
    android:textSize="18sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/regPageLogo" />

```

<EditText

```

    android:id="@+id/reg_confirm_pass"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:backgroundTint="@android:color/white"
    android:ems="10"
    android:hint="Confirm password"
    android:inputType="textPassword"
    android:padding="16dp"
    android:textColor="@android:color/white"
    android:textColorHint="#FFFFFF"
    android:textSize="18sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/reg_pass" />

```

<Button

```

    android:id="@+id/reg_btn"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:backgroundTint="@android:color/holo_red_dark"
    android:paddingBottom="20dp"
    android:paddingTop="20dp"
    android:text="Create new Account"
    android:textAllCaps="false"
    android:textColor="@android:color/white"
    android:textSize="18sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/reg_confirm_pass" />

```

<Button

```

    android:id="@+id/reg_login_btn"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginBottom="8dp"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:backgroundTint="@android:color/white"
    android:paddingBottom="20dp"
    android:paddingTop="20dp"
    android:text="Already have an Account"

```

```

        android:textAllCaps="false"
        android:textColor="#FF0000"
        android:textSize="18sp"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent" />

```

```

<ProgressBar
    android:id="@+id/reg_progress"
    style="?android:attr/progressBarStyleHorizontal"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:indeterminate="true"
    android:visibility="invisible"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toTopOf="parent" />

```

```

<EditText
    android:id="@+id/reg_pass"
    android:layout_width="0dp"
    android:layout_height="wrap_content"
    android:layout_marginEnd="8dp"
    android:layout_marginStart="8dp"
    android:layout_marginTop="8dp"
    android:backgroundTint="@android:color/white"
    android:ems="10"
    android:hint="Password"
    android:inputType="textPassword"
    android:padding="15dp"
    android:textColor="@android:color/white"
    android:textColorHint="#FFFFFF"
    android:textSize="18sp"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/reg_email" />

```

</android.support.constraint.ConstraintLayout>

Bước 4: Xử lý sự kiện trên LoginActivity

EditText **edt1**, **edt2**;

Button **btn1**, **btn2**;

@Override

```

protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_login);

```

```

    edt1 = findViewById(R.id.edtText);

```

```

    edt2 = findViewById(R.id.edtText2);

```

```

    btn1 = findViewById(R.id.button);

```

```

    btn2 = findViewById(R.id.button2);

    btn1.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            String name = edt1.getText().toString();
            String password = edt2.getText().toString();
        }
    });

    btn2.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent intent = new Intent(LoginActivity.this, RegisterActivity.class);
            startActivityForResult(intent, 100);
        }
    });
}

@Override
protected void onActivityResult(int requestCode, int resultCode, Intent data) {
    super.onActivityResult(requestCode, resultCode, data);
    if(requestCode==100 && resultCode==101){
        edt1.setText(data.getStringExtra("name"));
        edt2.setText(data.getStringExtra("pass"));
    }
}

```

Bước 5: Xử lý sự kiện trên RegisterActivity

EditText **edt1**, **edt2**, **edt3**;

Button **btn1**, **btn2**;

```

@Override
protected void onCreate(Bundle savedInstanceState) {
    super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_register);

    edt1 = findViewById(R.id.reg_email);
    edt2 = findViewById(R.id.reg_pass);
    edt3 = findViewById(R.id.reg_confirm_pass);

    btn1 = findViewById(R.id.reg_btn);
    btn2 = findViewById(R.id.reg_login_btn);

    btn1.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            String name = edt1.getText().toString();
            String pass = edt2.getText().toString();
            String conform = edt3.getText().toString();

            if(pass.compareToIgnoreCase(conform) == 0){
                Intent intent = new Intent(RegisterActivity.this, LoginActivity.class);
            }
        }
    });
}

```

```

        intent.putExtra("name", name);
        intent.putExtra("pass", pass);
        setResult(101,intent);
        finish();
        Toast.makeText(RegisterActivity.this, "Registered !!!", Toast.LENGTH_SHORT).show();
    }

    });

    btn2.setOnClickListener(new View.OnClickListener() {
        @Override
        public void onClick(View view) {
            Intent intent = new Intent(RegisterActivity.this, LoginActivity.class);
            startActivity(intent);
        }
    });
}

Bước 7: Xử lý AndroidManifest.xml để chọn activity khởi tạo
<activity android:name=".LoginActivity">
    <intent-filter>
        <action android:name="android.intent.action.MAIN" />

        <category android:name="android.intent.category.LAUNCHER" />
    </intent-filter>
</activity>

```